

S.M. Saad Hussain Zaidi

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[Portfolio](#)

Professional Summary

A highly motivated Physicist and Data Scientist with a strong focus on software engineering, machine learning, and IoT development. With consistent demonstration of tackling complex analytical problems, I build end-to-end data pipelines, predictive models, and intuitive web platforms. Whether deploying hybrid machine learning models to predict financial risk or developing full-stack solutions for hardware integration, I thrive at the intersection of data, software, and scientific computing to deliver actionable insights and scalable solutions.

Technical Skills

Machine Learning & AI: Deep Learning, Machine Learning Algorithms, Object Detection, Image Processing, Ultralytics (YOLOv8)

Data Engineering & Analytics: Python, SQL, Pandas, NumPy, Power BI, Data Wrangling, Statistics, Data Visualization

Web Development: React, HTML, CSS, JavaScript, Web Designing, Deployment

Hardware & IoT: Embedded Systems Engineering, PCB Designing, Hardware Prototyping, Firmware Development, Electronics

Professional Experience & Projects

Data Analyst & AI Intern | RESOLVE South, SUPARCO Complex.

(Jun 2025 – Aug 2025)

- **Solar Panel Detection & Change Detection AI Model:** Developed a YOLOv8-based detection and monitoring pipeline of solar panels from satellite imagery, enabling model comparison for accuracy and renewable-energy monitoring insights.

Data & Machine Learning Researcher | NED University.

(Dec 2024 – Jun 2025)

- **Bankruptcy Risk Prediction Using Hybrid ML Models:** Designed a Hybrid Dynamic Panel Probit + ANN (11-neuron single hidden layer, cross-entropy loss, k-fold CV) model using financial, macroeconomic, and market variables.
- **Flood Damage Prediction Model:** Created a two-stage classification pipeline (inundation depth + material type) using optimized Random Forest achieving 95% accuracy for minor damage and 83% for severe damage.

Machine Learning Intern | Pinnacle Full-Stack Interns.

(Jun 2024 – Jul 2024)

- **Email Spam Detection System:** Applied text preprocessing and implemented decision tree classification models for precise spam detection.
- **Heart Disease Classification:** Developed an ML classification pipeline tailored for healthcare and diagnostic analytics.

Data Analyst Intern | Placement Dost.

(Apr 2024 – Jun 2024)

- **Flight Activity Dashboard:** Built comprehensive Power BI dashboards to analyze booking trends, CLV segmentation, geographic distribution, and loyalty impact.

Education

- BS Physics | NED University of Engineering & Technology, Karachi.

(Oct 2022 – Dec 2026)

- Applied Data Science with Machine Learning | UK Tech Hub.

(Oct 2023 – Nov 2024)